

Bayes' Formula Questions

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Note: Questions are from Durrett, The Essentials of Probability.

To compute $P(B_1|A)$ when $P(B_i)$ and $P(A|B_i)$ are given, use the following three-step procedure:

$$P(B_1|A) = \frac{P(B_1 \cap A)}{P(A)}$$

by the definition of conditional probability. Then

$$P(B_i \cap A) = P(B_i)P(A|B_i).$$

Then

$$P(A) = \sum_{i=1}^n P(B_i \cap A).$$

Combining the three steps gives Bayes' formula:

Theorem (Bayes' Formula).

$$P(B_1|A) = \frac{P(B_1)P(A|B_1)}{\sum_i P(B_i)P(A|B_i)}.$$

Exercise 1. Suppose that the number of children in a family is 1, 2, or 3, with probability $1/3$ each. Little Bobby has no brothers. What is the probability he is an only child?

Exercise 2. Little Bobby has no sisters. What is the probability he is an only child?

Exercise 3. Binary digits are sent down a noisy communications channel. They are received as sent with probability 0.9 but errors occur with probability 0.1. Assuming that 0's and 1's are equally likely, what is the probability that a 1 was sent given that we received a 1?

Exercise 4. Two hunters shoot at a deer, which is hit by exactly one bullet. If the first hunter hits his targets with probability 0.3 and the second with probability 0.6, what is the probability the second hunter killed the deer? The answer is not $2/3$. Do you think the answer is larger or smaller?

Exercise 5. You are about to have an interview for Harvard Law School. 60 percent of the interviewers are conservative and 40 percent are liberal. 50 percent of the conservatives smoke cigars but only 25 percent of the liberals do. Your interviewer lights up a cigar. What is the probability he is a liberal?

Exercise 6. One slot machine pays off $1/2$ of the time, while another pays off $1/4$ of the time. We pick one of the machines and play it six times, winning 3 times. What is the probability we are playing the machine that pays off only $1/4$ of the time?

Exercise 7. 20 percent of people are accident-prone and have a probability 0.15 of having an accident in a one-year period in contrast to a probability of 0.05 for the other 80 percent of people. (a) If we pick a person at random, what is the probability he will have an accident this year?

(b) What is the probability a person is accident-prone if he had an accident last year?

(c) What is the probability he will have an accident this year if he had one last year?

Exercise 8. Three factories manufacture 20%, 30%, and 50% of the computer chips a company sells. If the probability of defective chips are 0.4%, 0.3%, and 0.2%, respectively, what is the probability that the defective chips come from the third factory?

Exercise 9. A company gives a test to 100 salesmen, 80 with good sales records and 20 with poor records. 60 percent of the good salesman pass the test, but only 30 percent of the poor salesmen do. A new applicant takes the test and passes. What is the probability he is a good salesman?

Exercise 10. Two masked robbers try to rob a crowded bank during the lunch hour but the teller presses a button that sets off an alarm and locks the front door. The robbers, realizing they are trapped, throw away their masks and disappear into the chaotic crowd. Confronted with 40 people claiming they are innocent, the police give everyone a lie detector test. Suppose that guilty people are detected with probability 0.95, and innocent people appear to be guilty with probability 0.01. What is the probability that Mr. Jones is guilty given that the lie detector says he is?